Flow studies on atriopulmonary and cavopulmonary connections of the Fontan operations for congenital heart defects.

PMID: 7689671 | Indexed in PubMed

A comparative flow study has been conducted on two configurations of the Fontan operations for congenital heart defects in which the right ventricle is by-passed. The study was made on rigid in vitro models of the atriopulmonary and cavopulmonary connections under steady-flow conditions. It involved using pressure and flow measurements to determine the pressure-drop coefficient, pressure-energy-loss coefficient and total-energy-loss coefficient. The cavopulmonary model was found to have much lower flow losses compared to the atriopulmonary model, especially at flow rates about 5 l min-1.

[Fontan associated liver disease].

PMID: 39219388 | Indexed in PubMed

Fontan surgery is vital for infants born with a single-ventricle heart. This intervention establishes a new blood flow circuit bypassing the single ventricle, thereby the separating pulmonary and systemic circulation to preserve single ventricular function. However, it carries risks of hepatic complications, collectively termed Fontan-associated liver disease (FALD), characterized by progressive hepatic congestion and fibrosis potentially leading to an equivalent of cirrhosis. Diagnosis and staging of FALD are complex, requiring multidisciplinary management. In advanced FALD, consideration is given to heart transplantation alone or combined heart-liver transplantation, underscoring the importance of an integrated approach to optimize care for these increasingly more common patients.

Modification of Hepatic Venous Conduit to Manage Pulmonary Arteriovenous Malformations.

PMID: 26180170 | Indexed in PubMed

While the Fontan operation is a reliable treatment option for many complex congenital heart defects, the development of pulmonary arteriovenous malformations (PAVMs) remains a problematic outcome for some Fontan patients. Pulmonary arteriovenous malformations stem from an imbalance of hepatic blood flow in the pulmonary system. Balancing this hepatic flow has shown promising results in the treatment of PAVMs. We report the clinical course of a young patient with heterotaxy syndrome and an unbalanced right dominant atrioventricular septal defect. This patient developed PAVMs following a Fontan procedure, however, the PAVMs were resolved following the revision of the original Fontan conduit to a bifurcated conduit.

[Comparison between patients undergoing Fontan operation with or without cardiopulmonary bypass].

PMID: 26830073 | Indexed in PubMed

Fontan operation is the final palliative stage of patients with univentricular hearts. Cardiopulmonary bypass (CPB) decreases ventricular performance and increases pulmonary artery pressures in the post operative recovery period. It seems that Fontan operation performed without CPB decreases short term morbidity and intra hospitalary length of stay. Compare outcome in Fontan patients who have undergone surgery with or without CPB. This is a retrospective review of patients undergoing Fontan operation from january 2009 to december 2012. Patients were grouped according to CPB use and comparative analyses were done. Ten patients were operated without CPB use. There was a discrepancy between age in both groups, being younger in the no CPB group. Around 80% of patients in both groups had a staged procedure. A 18mm graft was used in half of the cases; a fenestration was created in all cases. Length of stay was equal in both groups, there was less need of pharmacologic support and nitric oxide use in patients without CPB use. No deaths were reported also in this group. At folllow up, most patients had a class i functional status. In our experience, Fontan operation without CPB has similar outcomes compared with CPB use.

Success in a Fontan pregnancy: how important is ventricular function?

PMID: 30482258 | Indexed in PubMed

The Fontan operation is a palliative surgical procedure for patients whose hearts cannot support a biventricular circulation. The haemodynamic changes that occur in pregnancy are particularly challenging for Fontan patients and the outcomes are variable. We present a case where fetal outcome was particularly poor despite a lack of high risk features pre-pregnancy.

Outcome and assessment after the modified Fontan procedure for hypoplastic left heart syndrome.

PMID: 1728440 | Indexed in PubMed

We reviewed the outcome of 76 consecutive patients (age range, 5 months to 6 years; median age, 19 months) who underwent a modified Fontan procedure after initial palliative surgery for hypoplastic left heart syndrome (HLHS) between January 1984 and December 1989. Modifications of the Fontan procedure included transatrial baffle of pulmonary venous return to the tricuspid valve (n = 10) or inferior vena cava baffle within the right atrium to the superior vena caval-pulmonary artery anastomosis, with pulmonary artery augmentation (n = 66). Actuarial survival rates were 74% (1 month), 58% (12 months), 56% (2 years), and 52% (4 years). Of the 43 survivors, 25 patients have returned for postoperative cardiac catheterization at a medium of 13 months after the Fontan procedure. Mean +/- SD hemodynamic values were cardiac index, 2.8 +/- 0.6 l/min/m2; right arterial pressure, 11 +/- 2 mm Hg; pulmonary artery wedge pressure, 6 +/- 3 mm Hg; and arterial oxygen saturation, 94 +/- 3%. No patient had significant tricuspid or native pulmonary valve insufficiency. Survival after the Fontan procedure in patients with HLHS is comparable to survival after a Fontan procedure in patients with other complex congenital heart lesions. In the subgroup of patients with HLHS who survived both reconstructive surgery and a Fontan procedure and have been evaluated by cardiac catheterization after a Fontan procedure, the use of the right ventricle as the systemic ventricle yielded excellent intermediate results for Fontan physiology.

Fontan procedure for hypoplastic left heart syndrome.

PMID: 1280410 | Indexed in PubMed

Since 1985, 354 neonates have undergone palliative reconstruction for hypoplastic left heart syndrome with 109 early deaths and 12 late deaths. Of the survivors, before 1989, 77 patients underwent a subsequent modified Fontan operation, consisting of baffling the atrial septal defect to the tricuspid valve (initial 25 patients) or intraatrial baffling of the inferior vena cava to the pulmonary arteries and superior vena cava (52 patients). There were 17 early deaths and three late deaths. Major serous effusions developed in 42 patients (54%) after Fontan operation. Since 1989, a staged approach to Fontan's operation was undertaken in an effort to reduce the volume load of the right ventricle as early as possible, to minimize the impact of rapid changes in ventricular geometry and diastolic function that can accompany a primary Fontan operation, and to reduce effusive complications. Thus, at a mean age of 6 months, 121 patients have undergone closure of aortopulmonary shunt, augmentation of central pulmonary arteries, and association of the superior vena cava with the branch pulmonary arteries (hemi-Fontan procedure). Of these, 61 patients have already undergone completion of the Fontan procedure with six early deaths and three late deaths. Major serous effusions developed in 28 patients (46%) with the staged Fontan. For perspective, the contemporary experience since January 1991 consists of 58 neonates who have undergone initial palliation with 11 deaths (19%), 17 patients who have undergone the hemi-Fontan procedure with one death (6%), and 21 patients who have undergone completion of the Fontan operation with one death (5%).(ABSTRACT TRUNCATED AT 250 WORDS)

The hemi-Fontan operation.

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In hearts with a functional single ventricle, cavity volume and myocardial muscle mass increase as a consequence of the excessive volume load associated with parallel pulmonary and systemic circulations. The hemi-Fontan operation was conceived as a means of accomplishing early reduction of the volume work of the single ventricle, in anticipation of eventual completion of a modified Fontan procedure. The hemi-Fontan procedure includes association of the superior vena(e) cava(e) (SVC) with the branch pulmonary arteries, augmentation of the central pulmonary arteries, occlusion of the inflow of the SVC into the right atrium, and elimination of other sources of pulmonary blood flow. Because it is preparatory to an eventual completion Fontan procedure, concomitant procedures are conveniently performed in conjunction with the hemi-Fontan. These may include atrial septectomy, relief of aortic arch obstruction, proximal main pulmonary artery to ascending aortic anastomosis, repair or revision of anomalous pulmonary venous connection, atrioventricular valvuloplasty, or other procedures.

Phase 2 Trial of Anti-TL1A Monoclonal Antibody Tulisokibart for Ulcerative Colitis.

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Tulisokibart is a tumor necrosis factor-like cytokine 1A (TL1A) monoclonal antibody in development for the treatment of moderately to severely active ulcerative colitis. A genetic-based diagnostic test was designed to identify patients with an increased likelihood of response. We randomly assigned patients with glucocorticoid dependence or failure of conventional or advanced therapies for ulcerative colitis to receive intravenous tulisokibart (1000 mg on day 1 and 500 mg at weeks 2, 6, and 10) or placebo. Cohort 1 included patients regardless of status with respect to the test for likelihood of response. Cohort 2 included only patients with a positive test for likelihood of response. The primary analysis was performed in cohort 1; the primary end point was clinical remission at week 12. Patients with a positive test for likelihood of response from cohorts 1 and 2 were combined in prespecified analyses. In cohort 1, a total of 135 patients underwent randomization. A significantly higher percentage of patients who received tulisokibart had clinical remission than those who received placebo (26% vs. 1%; difference, 25 percentage points; 95% confidence interval [CI], 14 to 37; $P<0.001$). In cohort 2, a total of 43 patients underwent randomization. A total of 75 patients with a positive test for likelihood of response underwent randomization across both cohorts. Among patients with a positive test for likelihood of response (cohorts 1 and 2 combined), clinical remission occurred in a higher percentage of patients who received tulisokibart than in those who received placebo (32% vs. 11%; difference, 21 percentage points; 95% CI, 2 to 38; $P = 0.02$). Among all the enrolled patients, the incidence of adverse events was similar in the tulisokibart and placebo groups; most adverse events were mild to moderate in severity. In this short-term trial, tulisokibart was more effective than placebo in inducing clinical remission in patients with moderately to severely active ulcerative colitis. (Funded by Prometheus Biosciences, a subsidiary of Merck; ARTEMIS-UC ClinicalTrials.gov number, NCT04996797.).

A flow-cytometry-based assay to assess granule exocytosis and GZB delivery by human CD8 T cells and NK cells.

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CD8 T and NK cells mediate killing by delivery of perforin and granzyme B (GZB) stored in lysosome-like granules. We present a flow-cytometry-based protocol combined with a redirected killing assay to evaluate granule exocytosis and the cytotoxic potential of human CD8 T cells and NK cells. We describe the assessment of the delivered GZB inside the target cells. We then detail the detection of lysosome membrane protein CD107a exposed on the cell surface of the effector cells upon degranulation. For complete details on the use and execution of this protocol, please refer to Chen et al. (2021).1.